

CS 301 Midterm — S003 — Question 1

QuickSort uses a divide-and-conquer approach.

In the average case, it takes $O(n \log n)$ time because the pivot splits the array into two r

The worst case happens when the pivot is always the maximum or minimum element, giving $O(n^2)$

The partition step takes $O(n)$ time.

CS 301 Midterm — S003 — Question 2

BFS stands for Breadth First Search.

It explores the graph layer by layer using a queue.

DFS stands for Depth First Search and uses a stack or recursion.

BFS is great for shortest path problems.

They both take $O(V + E)$ time.